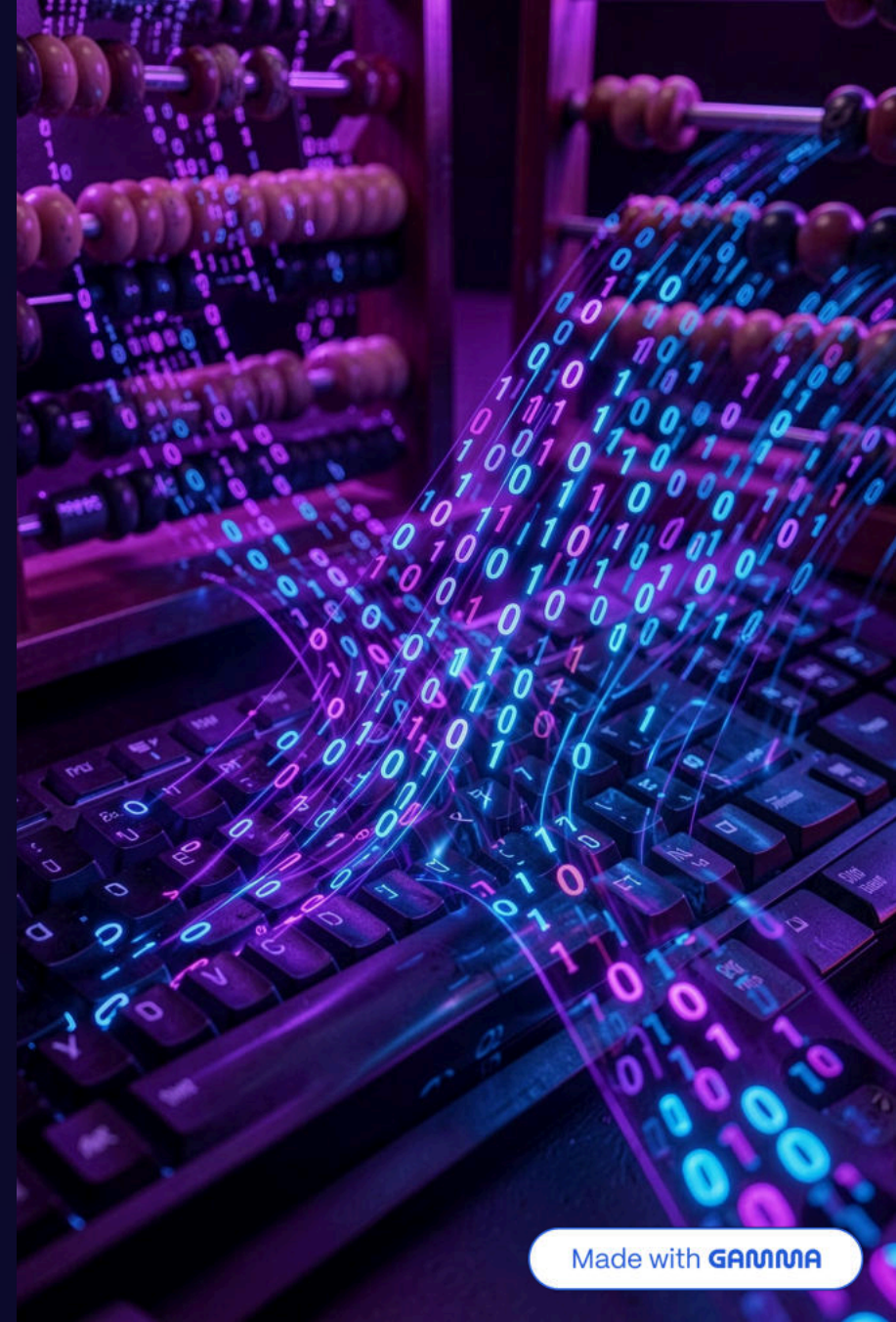


Number Systems & Binary Numbers

GRADE 10 · ICT · MODULE 3.1

ICT Grade 10 - Chapter 3 - Module 1 - Akila Savinda



What Will We Learn Today?



How Computers Represent Data

Understand how computers process and store information using numbers



Four Number Systems

Identify Binary, Octal, Decimal, and Hexadecimal systems and their uses



Why Binary?

Explain why computers rely on just two digits — 0 and 1



Key Vocabulary

Define Bit, Byte, Nibble, Base/Radix, MSB, and LSB

How Does a Computer Understand Data?



The Journey of the Letter "A"

Chanaka and his friends wondered:
*"When 'A' is pressed on the keyboard,
how does the computer identify it?"*



When typing letters or words, the computer represents them as numbers it can understand.

01

Press "A" on the keyboard

02

Electronic signal sent to the System Unit

03

Signal converted to ASCII Binary Code: **1000001**

04

Data passes through RAM to video memory

05

"A" is displayed on the screen

What is a Number System?

A **Number System** is the group of numbers that a computer can understand. It is defined by three key properties:



The concept dates back to the **Abacus** — the world's first calculating machine — and has evolved into today's computers.



Digit

The limited set of numerals used in the system

Positional Value

A number's value depends on its position within the number

Base / Radix

The total number of symbols used in the system

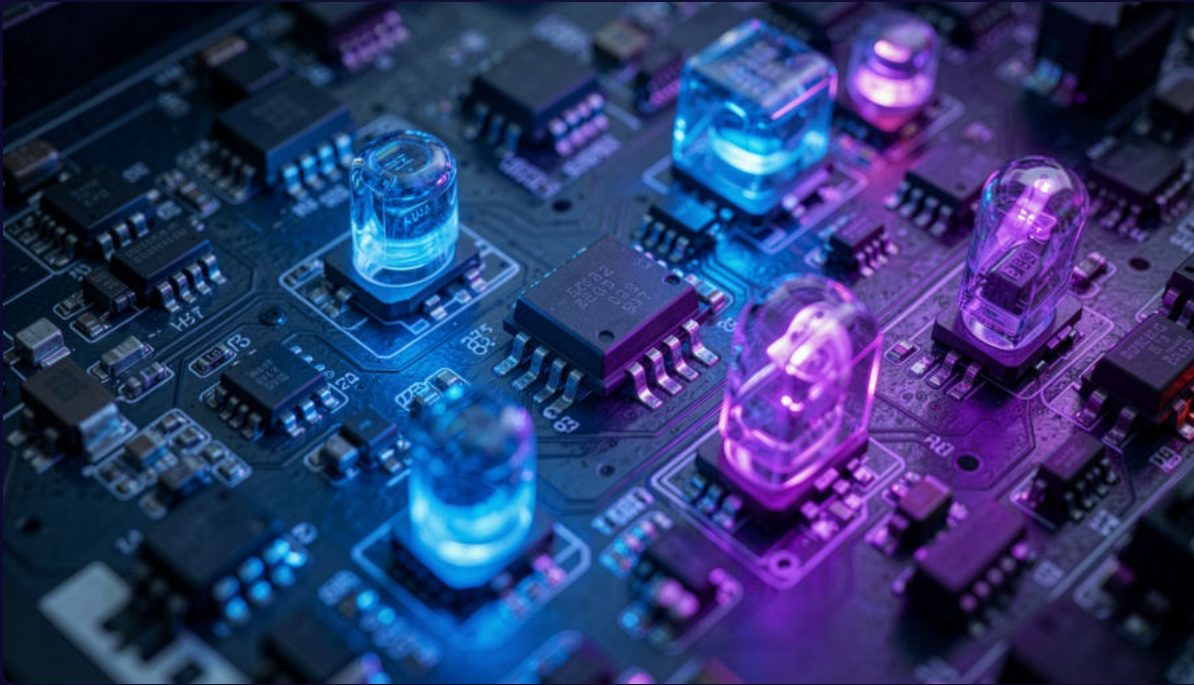
The Four Number Systems

Every number system is defined by its **base** — the number of unique symbols it uses.

Number System	Base	Digits / Symbols Used
Binary	2	0, 1
Octal	8	0, 1, 2, 3, 4, 5, 6, 7
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Hexadecimal	16	0–9, A, B, C, D, E, F

 **Key Rule:** The base of any number system is always expressed in decimal. Base 2 → 2 digits; Base 10 → 10 digits.

Why Binary? The Language of Electronics



Two Voltage Levels

Computers use electronic circuits that exist in only **two states**:

Any data in the world — text, images, sound, video — can be represented using just these two digits.



Real-World Example —

Colours: Every screen colour uses RGB values (0–255). Dark Purple = R:135, G:31, B:120 → Binary: 10000111, 00011111, 01111000 → Hex: **#871F78**

High Voltage → "1"

On state of the circuit

Low Voltage → "0"

Off state of the circuit

Binary in Detail & The Bit

Binary Number System

Base 2, using digits **0** and **1**. Example: **11101101₂**

Each position has a **weighting factor** (powers of 2):

1	1	1	0	1	1	0	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

The Bit — Smallest Unit

Bit = **B**inary digit. The most basic unit of data in computing.

1 Bit

2 values (0 or 1)

4 Bits = 1 Nibble

16 possibilities

8 Bits = 1 Byte

256 possibilities

Octal & Hexadecimal Number Systems

Octal (Base 8)

Digits: **0–7** · Example: 236_8

Weighting factors: $8^0, 8^1, 8^2 \dots$

$$236_8 = (2 \times 64) + (3 \times 8) + (6 \times 1) = 158_{10}$$

Key link: $8 = 2^3 \rightarrow$ each octal digit = **3 binary bits**

Hexadecimal (Base 16)

Digits: **0–9, A–F** · Example: $15E_{16}$

Used because binary is hard for humans to read. Hex is compact and easy to convert.

$$15E_{16} = (1 \times 256) + (5 \times 16) + (14 \times 1) = 350_{10}$$

Decimal	Hex
10	A
11	B
12	C
13	D
14	E
15	F

MSD, LSD, MSB & LSB

Most & Least Significant *Digit*

Applies to **any** number system. Read left to right:

- **MSD** = Leftmost *non-zero* digit
- **LSD** = Rightmost digit (including after decimal)

Number	MSD	LSD
329	3	9
1237.01	1	1
20.0975	2	5

Most & Least Significant *Bit*

Applies to **binary only**:

- **MSB** = Leftmost **1** (highest power of 2)
- **LSB** = Rightmost digit (lowest power of 2)

Binary	MSB	LSB
1001	1 (2^3)	1 (2^0)
011.101	1 (2^1)	1 (2^{-3})



Activity: Find the MSB and LSB of: (i) 1000, (ii) 011101, (iii) 0.11001, (iv) 1.0010, (v) 0.00110

Module 3.1 — Summary & Review

Key Takeaways

→ Four Number Systems

Binary (2), Octal (8), Decimal (10), Hexadecimal (16)

→ Why Binary?

Computers use two voltage levels — High (1) and Low (0)
— representing On/Off circuit states

→ Key Terms

Bit = binary digit · **Byte** = 8 bits · **MSB/LSB** = most/least significant bit · **MSD/LSD** = most/least significant digit

Review Questions

1. Why do computers use binary numbers?
2. What is the base of the octal number system?
3. How many digits does hexadecimal use?
4. What is a bit?
5. Explain the difference between MSB and LSB.
6. Using the conversion table, express decimal **10** in binary, octal, and hexadecimal.



Use Table 3.8 to verify your conversions for question 6!